Shaownak Shahriar

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Experience

AI & Computer Vision Engineer — May 2026 – Present
Roboway Technologies — Dhaka, Bangladesh

- Developed **enterprise-scale computer vision infrastructure** and workforce intelligence systems for **Banglalink**.
- Designing **scalable AI pipelines** and deployment workflows for production environments.

Autonomous Systems Engineer — May 2024 – Apr 2025
Dubotech Ltd. — Dhaka, Bangladesh

- Engineered and deployed real-time underwater image enhancement pipelines using **GAN-based translation** to process feeds in low-visibility environments
- Optimized inference latency by deploying vision models onto **NVIDIA Jetson Orin Nano** utilizing **TensorRT**, achieving stable performance for live video streams
- Integrated **Pixhawk, Raspberry Pi, and Jetson based** autonomous control systems with onboard sensors.

AI & Autonomous Systems Lead — Jun 2021 – Apr 2024
BRACU Duburi & Mongol-Tori — Dhaka, Bangladesh

- Led a **30+ engineer team** developing AI and navigation pipelines for AUV and Mars rovers, securing **Global 1st Runner-up (RoboSub 2023)** and **Global 7th Place (IRC 2023)**.
- Developed robust perception and vision pipelines utilizing **YOLO, Faster/Mask R-CNN, SAM, and U2-Net** along with **stereo vision** and **monocular depth estimation** for real-time target tracking and segmentation.
- Implemented **Inverse Kinematics** for a multi-DOF robotic arm and tuned closed-loop **PID control systems** for precise actuator and motor driver stabilization.
- Designed real-time state estimation and localization pipelines using **ROS-based SLAM** and sensor fusion (**Kalman/Madgwick filters**) across LiDAR, GPS, IMU, and RealSense.

Key Projects

Point-Cloud-Based 3D Object Detection for UGVs | *Best Thesis Award 2024*

- Proposed a hybrid **GLE-SSD-VR voxel-point architecture** for LiDAR-based 3D object detection, optimizing real-time point-cloud processing.
- Evaluated and benchmarked performance using the **KITTI dataset** for robust ground vehicle navigation.

BRACU RaptorX — Autonomous VTOL UAV | *Co-Founder & Autonomy Lead*

- Co-founded development of **Bangladesh's first autonomous quad-plane VTOL UAV**, integrating sensor fusion, AI-based flight control, and real-time computer vision.
- Led autonomy and navigation subsystem development to handle state estimation, path planning, and fully autonomous mission execution.
- **Operationally deployed** the platform during flood-relief operations in Feni for emergency supply delivery, validating system reliability in harsh environments.

Driver Safety & Road Perception System | *YOLOPv2, MiDaS, Jetson Nano*

- Developed a real-time multi-task driver monitoring system combining **lane detection, drowsiness monitoring, and monocular depth estimation**.

- Optimized deep learning model architectures to run efficiently on resource-constrained edge hardware (**NVIDIA Jetson Nano**).

Drone Detection System | *YOLOv11, Raspberry Pi*

- Fine-tuned and deployed a **YOLOv11 pipeline** for real-time target detection and tracking on embedded hardware.
- Implemented custom edge-inference optimization techniques to maximize frame rate and minimize latency on a **Raspberry Pi**.

Geospatial Traffic Optimization System | *Geospatial AI, GNNs, LSTMs, XGBoost, OSMnx, Folium*

- Developed a spatial-temporal traffic prediction framework for Dhaka's road networks using **Graph Neural Networks (GNNs)** and ensemble models to optimize **routing efficiency and supply chain logistics**.

Global Achievements

1st Runner-up (Global) — RoboSub 2023, RoboNation (California, USA)

7th Place (Global) — International Rover Challenge 2023, (Bangalore, India)

Participant — University Rover Challenge 2023 (Utah, USA)

Participant — IMechE UAS Challenge 2024 (Leciester, England)

3rd Place — Kibo Robot Programming Competition 2022, JAXA (Dhaka, Bangladesh)

Technical Skills

Languages & Core Tools: C/C++, Python, SQL, Bash, Linux, Git, Docker, CVAT
Robotics Frameworks: ROS, Gazebo, RViz
Autonomy & Navigation: SLAM, Navigation Stack, Path Planning, State Estimation, Kalman Filter, Madgwick Filter, Flight Control
Control & Kinematics: Forward/Inverse Kinematics, PID Control
Computer Vision & AI: TensorFlow, PyTorch, OpenCV, TensorRT, ONNX, YOLO, SSD, GANs, VectorDB
Communication & Protocols: UART, SPI, I2C, CAN Bus, MAVLink, MQTT Broker
Sensors & Hardware: LiDAR, Intel RealSense, IMU, GPS/RTK, VectorNav, SBG INS, Pixhawk, Jetson Orin/Nano, Arduino, Motor Drivers(Cytron, IBT)

Education

BRAC University Graduated 2024
Bachelor of Science in **Computer Science** Dhaka, Bangladesh

- **Best Thesis Award 2024** (Computer Science Department)
- **CGPA:** 3.14 / 4.00

Mentorship & Instruction

Guest Robotics Instructor 2025
Pabna University of Science and Technology Pabna, Bangladesh

- Delivered advanced practical sessions on **robotics, automation, and control systems** for undergraduate engineering students.

Robotics Instructor Jan 2020 – Aug 2023
BRAC University & Robotics Club of BRACU (ROBU) Dhaka, Bangladesh

- Taught **Basics of Robotics** in BRACU Residential Semester.
- Conducted specialized workshops on **Computer Vision and Autonomous Systems** to train junior engineers in Robotics Club.

ENGINEERING PORTFOLIO

SHAOWNAK SHAHRIAR

Robotics, Computer Vision, Autonomous Systems

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About Me

I am a **Robotics and Artificial Intelligence Engineer** passionate about designing **intelligent autonomous systems** that solve real-world industrial and societal challenges. My expertise spans **robotics, computer vision, artificial intelligence, autonomous navigation, embedded systems, control systems, and industrial automation,** with a strong focus on developing end-to-end intelligent robotic solutions.

Throughout my academic and professional journey, I have contributed to and led multidisciplinary engineering projects, including the development of Bangladesh's first autonomous quad-plane **VTOL UAV**, **autonomous underwater vehicles (AUVs)**, **autonomous Mars rovers**, **AI-powered rescue robots**, **enterprise computer vision systems**, and **advanced perception solutions**. My work integrates **AI algorithms, embedded computing, sensors, and control systems** to build complete autonomous platforms—from perception and decision-making to navigation, control, and deployment.

My technical expertise includes **Computer Vision, Deep Learning, Robotics, Autonomous Systems, Embedded AI, Sensor Fusion, and Intelligent Control Systems.** I enjoy transforming research ideas into practical engineering solutions by combining artificial intelligence, robotics, and automation technologies to solve complex real-world problems.

Beyond technical development, I have experience **leading multidisciplinary engineering teams**, managing complex robotics projects, **mentoring junior engineers**, and **collaborating across diverse technical domains** to deliver innovative solutions under real-world constraints.

Driven by continuous learning and innovation, I aspire to **contribute to the next generation of intelligent robotic systems** for industrial automation, autonomous inspection, manufacturing, logistics, and smart infrastructure. My goal is to develop AI-powered robotic technologies that improve safety, efficiency, reliability, and productivity while creating meaningful impact across industries.

ENGINEERING PROFILE

Core Specializations

- Computer Vision
- Autonomous Systems
- Robotic Control System
- Artificial Intelligence
- Embedded AI
- UAV Autonomy
- Industrial Automation
- Robotics Software Engineering
- Sensor Fusion
- Edge AI Deployment
- Real-Time Embedded Systems

ENGINEERING HIGHLIGHTS

Domain Experience:

- **Computer Vision:** Object Detection/Segmentation, Image Processing, Industrial Machine Vision, Underwater Vision, 3D Perception
- **Robotics:** UAVs, Industrial Robots, Mobile Robotics, Autonomous Navigation
- **Artificial Intelligence:** Deep Learning, CNNs, Transfer Learning, Edge AI, Transformars, GANs
- **Embedded Systems:** Jetson, Raspberry Pi, Pixhawk, STM32, ESP32, Arduino
- **Software Engineering:** Python, C++, OpenCV, PyTorch, ROS, Linux
- **Industrial Systems:** Industrial fault/defect detection, Enterprise grade machine vision system deployment
- **Research Interest:** 3D Object Detection, Underwater Vision Enhancement, UAV Autonomy, Imitation Learning for Robotic Control

Engineering Highlights

- Co-founded and led the Autonomous & Navigation sub-team of **BRACU RaptorX**.
- Contributed to the development of Bangladesh's first autonomous quad-plane **VTOL UAV**.
- Built AI-powered perception systems using modern computer vision techniques.
- Worked on **autonomous underwater robotics** and **rescue robotics** projects.
- Experience integrating software, embedded hardware, sensors, and autonomous control systems.
- Currently working in **enterprise IT** while pursuing advanced robotics and industrial automation opportunities.

Career Objective

To contribute as a Robotics Engineer by developing intelligent autonomous systems that combine robotics, artificial intelligence, computer vision, and industrial automation. My goal is to transform innovative ideas into reliable engineering solutions that improve efficiency, safety, and productivity in real-world industrial environments while continuously expanding my expertise in advanced robotics and intelligent automation.

Computer Vision Based Employee Management System

Organization: Roboway Technologies
Client: Banglalink Digital Communications Ltd.
Role: AI & Computer Vision Engineer
Project Type: Enterprise AI Surveillance & Workforce Management Platform

Project Overview

Designed and developed a **production-grade** AI-powered **Vision-Based Employee Management System** for **intelligent workforce monitoring** and security automation. The platform leverages real-time computer vision and deep learning to process multiple live camera streams simultaneously, enabling **automated employee identification, attendance management, behaviour detection, security monitoring, and intelligent event notification**. Built for enterprise-scale deployment, the system enhances operational efficiency, strengthens workplace security, and significantly reduces manual administrative overhead.

Core Technologies

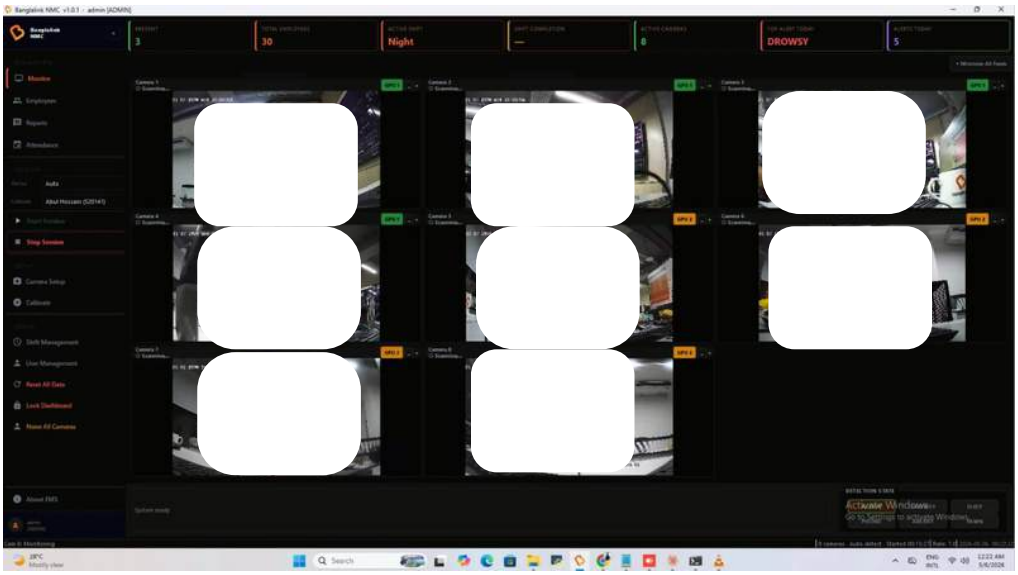
- **Computer Vision:** YOLO, MediaPipe, FaceNet, MRTCNN, FAISS, OpenCV, PyTorch, TensorRT
- **Backend:** Django, REST API, Kafka, Docker, Ngnix
- **Database:** MongoDB, Redis
- **Programming Languages:** Python
- **Deployment Environment:** Windows
- **Notification Service:** Telegram Bot API

My Responsibilities

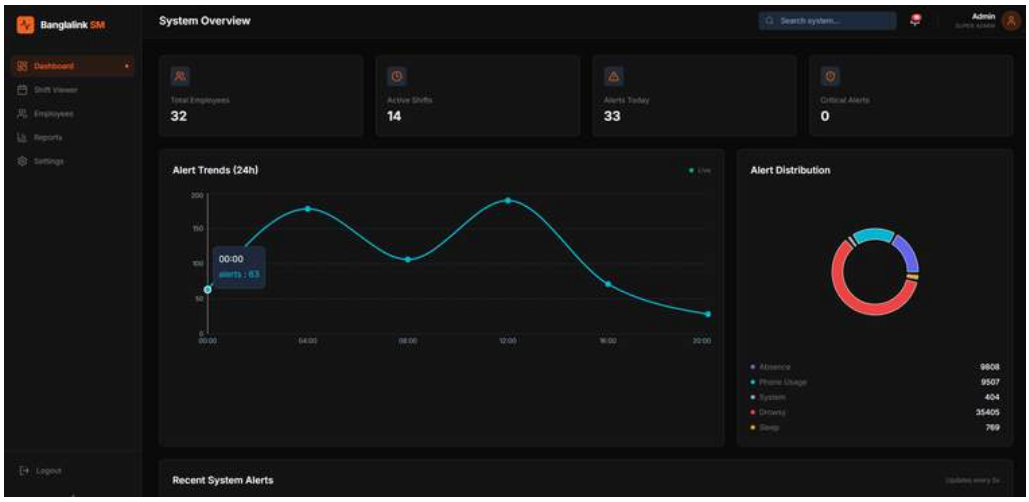
- Architected and developed the end-to-end AI vision pipeline for enterprise-scale deployment.
- Engineered a real-time analytics engine capable of processing 10 simultaneous live camera streams.
- Implemented face detection, face recognition, liveness detection, and intelligent employee identification.
- Designed and integrated automated attendance tracking and behaviour detection.
- Developed intelligent alerting mechanisms with instant Telegram-based notifications for security events.
- Integrated AI modules with the enterprise dashboard, backend services, and database infrastructure.
- Optimized inference performance, system scalability, and real-time reliability for continuous operation.

Key Features

- Real-time processing of 10 concurrent IP camera streams
- AI-powered employee recognition and attendance automation
- Face anti-spoofing (liveness detection)
- Behaviour, absense, sleep, phone usage detection
- Unknown person detection with instant security alerts
- Live monitoring dashboard with analytics and reporting
- Real-time Telegram notification system
- Scalable enterprise-ready architecture
- High-performance, low-latency AI inference



Detection UI



Web Dashboard

BRACU RaptorX – Autonomous Quad-Plane VTOL UAV

Co-Founder | Autonomous & Navigation Lead

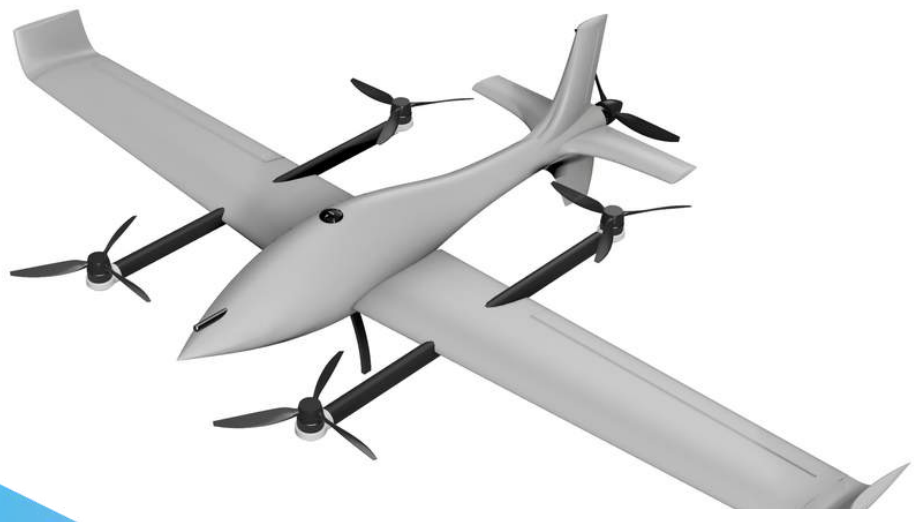
Project Type: Autonomous Unmanned Aerial Vehicle (VTOL)
Duration: December 2023 - December 2024
Organization: BRAC University, LASSET

Project Overview

The BRACU RaptorX project focused on the research, design, and development of Bangladesh's first autonomous quad-plane **Vertical Take-Off and Landing (VTOL)** unmanned aerial vehicle. The aircraft was engineered to combine the vertical take-off capability of a multirotor with the long-range efficiency of a fixed-wing aircraft, enabling fully autonomous missions in challenging environments.
Designed entirely in-house by a multidisciplinary student engineering team, the platform supports **autonomous navigation, waypoint-based mission execution, and real-time telemetry**. The project demonstrates the integration of mechanical design, embedded systems, flight control, autonomous software, and field deployment into a complete aerial robotics solution.







Techical Specifications

- **Aircraft Type:** Autonomous Quad-Plane VTOL UAV
- **Flight Controller:** Pixhawk 6X
- **Companion Computer:** Raspberry Pi 4 Model B
- **Autopilot:** ArduPilot
- **Navigation System:** Radiolink M8N SE100 GNSS with Integrated Compass
- **Ground Control Station:** Mission Planner, QGroundControl
- **Communication Protocol:** MAVLink
- **Programming Languages:** Python, C++
- **Operating System:** Ubuntu Linux
- **Autonomous Capabilities:** Autonomous Take-off & Landing, Waypoint Navigation, VTOL Transition, Return-to-Launch (RTL), Helipad/Crowd Detection, Tracking
- **Vision:** SSD-MobileNet, ONNX, TensorFlow-Lite



PixHawk 6X



Telemetry Module



Raspberry Pi 4B



Radiolink SE100 GPS Module

My Responsibilities

- **Led** the **Autonomous & Navigation sub-team** throughout the design, development, and testing phases of the project.
- Designed and **implemented core control system and autonomous flight capabilities** using the ArduPilot ecosystem.
- **Configured and tuned the Pixhawk 6X** flight controller for stable VTOL and fixed-wing operations.
- **Integrated the Raspberry Pi 4 Model B** as the companion computer for onboard processing and mission execution.
- **Configured the GNSS, Telemetry** module to enable accurate positioning and autonomous waypoint navigation and control.
- Planned, executed, and validated autonomous missions using Mission Planner, QGroundControl and the MAVLink communication protocol.
- **Integrated onboard hardware components**, including sensors, telemetry systems, and flight control electronics.
- Performed **system debugging, sensor calibration, and flight controller parameter** tuning to improve reliability and flight performance.
- **Conducted field testing**, analyzed flight logs, and optimized autonomous navigation performance based on real-world test results.
- Collaborated with the mechanical, avionics, and electronics teams to ensure seamless integration of the complete VTOL platform.

Key Achievements

- Successfully **deployed** during the **Feni Flood Relief Operation** to support humanitarian assistance through aerial reconnaissance and situational awareness.
- Participated in the **IMechE UAS Challenge 2024** in Leceister, England.
- Organized and **conducted technical seminars** on autonomous UAV systems, robotics, and aerospace engineering.


UAS CHALLENGE
INSTITUTION OF MECHANICAL ENGINEERS

THIS IS A CERTIFICATE OF PARTICIPATION AWARDED TO

SHAOWNAK MD IBNE SHAHRIAR TALUKDER
BRAC UNIVERSITY

AT THE

UAS CHALLENGE 2024

HELD ON

1 JULY – 4 JULY 2024

PRESENTED BY THE

INSTITUTION OF MECHANICAL ENGINEERS

PAUL LLOYD
CHAIR OF THE UAS CHALLENGE
INSTITUTION OF MECHANICAL ENGINEERS

IMechE UAS Certificate

Photos and Media Coverage

উদ্যোগ

বন্যার্ত এলাকায় ড্রোন নিয়ে

মো. জান্নাতুল নাঈম

দলটির নাম র‍্যাপটরএক্স। বন্যাকবলিত ফেনীতে ড্রোন নিয়ে হাজির হয়েছিল ব্র্যাক ইউনিভার্সিটির শিক্ষার্থীদের এই দল। ড্রোনের সাহায্যে ত্রাণ দেওয়ার পাশাপাশি বিভিন্ন এলাকার অবস্থা পর্যবেক্ষণ করেছে তারা।

রোবোটিকস নিয়ে বিশ্ববিদ্যালয়ে ছোট-বড় প্রকল্প হলেও ড্রোন নিয়ে উল্লেখযোগ্য তেমন কাজ হয়নি। এই ভাবনা থেকেই কাজ শুরু করেন একদল শিক্ষার্থী—মোহাম্মদ শায়েখ সেলিম (দলনেতা), সৈয়দ জয় (সহ-দলনেতা), শুভনক শাহরিয়ার, ফসিউল আবেদীন খান, ফারদিন নওশাদ, মাইদুল ইসলাম, নাশওয়াহ আফিফ, তাজওয়ার আহমেদ, তাসনিয়া মেহজাবীন, প্রীতম দত্ত ও আবরার জাহিন। পরামর্শক ও উপদেষ্টা হিসেবে সঙ্গে ছিলেন গবেষণা সহকারী ফিরোজ ওয়াদুদ ও প্রভাষক আব্দুল্লাহ হিল কাফি।

গত বছর থেকেই প্রকল্পের খুঁটিনাটি নিয়ে কাজ চলছিল। বিস্তারিত জানালেন সৈয়দ জয়। তিনি বলেন, '২০২৩ সাল থেকে আমরা গবেষণা ও উন্নয়নের কাজ শুরু করি। ড্রোনের ক্ষমতা নিয়ে গবেষণা করার সময় বুঝতে পারি, বন্যা আক্রান্ত মানুষের সাহায্যে কাজ করতে পারে এই ড্রোন। যেখানে নৌকা পৌঁছানো সম্ভব নয়, সেখানেও আমরা ত্রাণ ও প্রয়োজনীয় জিনিস পাঠাতে পারি। তাই দেশের এই দুর্যোগের সময়ে প্রযুক্তিগত সাহায্য নিয়ে এগিয়ে আসার সিদ্ধান্ত নিই। বিশ্ববিদ্যালয় কর্তৃপক্ষও আমাদের সাহায্য করে।'

গত ২৭ থেকে ২৯ আগস্ট ফেনীতে কাজ করেছে র‍্যাপটরএক্স। এই উদ্যোগকে সফল দাবি করে দলনেতা মোহাম্মদ শায়েখ সরেজমিন কাজ করার অভিজ্ঞতা জানালেন। তাঁর ভাষ্য, 'আমরা প্রথমে ফেনীর ছাগলনাইয়া থেকে শুরু করে দাগনভূঞা ও সিলোনিয়া পর্যন্ত এলাকাগুলো নজরদারি করেছি। ড্রোনের মাধ্যমে এরপর দুর্গম এলাকায় ত্রাণ বিতরণ করি। মূলত দুটি ড্রোন ব্যবহার করেছি। ভিটিওএল (ভার্টিক্যাল টেক অফ অ্যান্ড ল্যান্ডিং) ড্রোন দিয়ে বিভিন্ন এলাকা পর্যবেক্ষণ করেছি। কোথায় নৌকা পৌঁছানো সম্ভব নয়, বোঝার চেষ্টা করেছি। আর কোয়াড ড্রোন দিয়ে চিহ্নিত এলাকায় প্রয়োজনীয় জিনিস ড্রপ করেছি। এর মধ্যে শুকনা খাবার, ওষুধ ও স্যানিটারি প্যাড ছিল।'

ছোট আকারের প্লেনের মতো দেখতে ভিটিওএল ড্রোনটি অল্প জায়গা ব্যবহার করে খাড়াভাবে আকাশে উড়তে সক্ষম। এটি দীর্ঘক্ষণ আকাশে উড়ে বিভিন্ন তথ্য পেতে সাহায্য করে। প্রায় ১ কিলোমিটার এলাকাজুড়ে স্বয়ংক্রিয়ভাবে কাজ করতে পারে, যা যেকোনো দুর্যোগ পরিস্থিতিতে ভূমিকা রাখতে বেশ উপযোগী।

এ ধরনের ড্রোন নজরদারি, মানচিত্র তৈরি, ত্রাণ বিতরণ, কৃষিকাজ এবং নিরাপত্তার কাজগুলোতে বিশেষভাবে কার্যকর বলে দাবি করছে র‍্যাপটরএক্স।



দুই ধরনের ড্রোন নিয়ে ফেনীর কন্যাকবলিত এলাকায় গিয়েছেন ব্র্যাকের এই শিক্ষার্থীরা। ছবি : সংগৃহীত

The Daily Star

BRACU drone team – RaptorX – conducts relief operations in Feni

FATIMA ASHRAF

BRACU RaptorX, a drone team from BRAC University (BRACU) has successfully completed its mission of delivering relief materials like medicines, sanitary napkins, and dry food items to several remote flood-affected areas in Feni.

The recent floods devastated much of Feni, with six upazilas—Chhagalnaiya, Fulgazi, Parshuram, Feni Sadar, Daganbhuiyan, and Sonagazi – completely submerged and nearly cut off from receiving relief materials due to the lack of boats, according to Feni district administration sources.

"Recognising the need for a more equitable distribution of aid, we started surveillance in the remote Chhagalnaiya Upazila and made our way to Daganbhuiyan and Silonia, where people were in dire need of relief materials," said Mohammad Shayekh, the team leader of RaptorX. The team used Light Detection and Ranging (LiDAR) technology for smooth surveillance and deployed quad drones to reach smaller, isolated communities to translate their words into action.

RaptorX is a cutting-edge VTOL (Vertical Take-Off and Landing) drone, combining the best features of both multirotor and fixed-wing aircraft. It boasts a hybrid propulsion system, utilising four propellers and a rear thruster, making it far more energy-efficient than standard multirotor drones. The thruster allows for



PHOTO: COURTESY

quick take-offs, and the transition to horizontal flight conserves power during longer missions. This extended flight time is critical for long-distance operations like the ones they have undertaken in Feni.

Equipped with AI, GPS, and sensors, RaptorX can autonomously navigate multifarious environments, avoiding obstacles in real time. Its versatility in carrying various payloads – from cameras and sensors to specialised cargo – means it can perform surveillance, mapping, and environmental monitoring in a single flight. Built to endure harsh weather conditions, RaptorX excels even in wind and rain, making it a reliable solution for critical missions like disaster relief.

The drone is the result of tireless efforts by a talented team of eleven students from BRACU. Their advisor, Abdulla Hil Kafi, a lecturer at BRACU's Electrical and Electronic Engineering (EEE) department, along with their mentor, Firoz Wadud, a research assistant at BRACU's Department of Computer Science and Engineering (CSE), guided the team throughout its journey.

RaptorX is divided into several sub-teams, each responsible for a critical aspect of the drone's development. Mechanical team lead Mohammad Shayekh and autonomous and navigation lead Shawnak Shahriar ensured the drone's mechanics and autonomous capabilities worked seamlessly. Syed Joy, leading the software and control team, built robust systems for communication and flight control. Network and communication lead Mohammad Fasiul Abedin Khan, and avionics lead Farrdin Nowshad, enhanced the drone's connectivity and onboard electronics. Design and simulation were handled by Maidul Islam Moon and his team member, Abrar Zahin, while outreach efforts were spearheaded by Nashowah Afif Nuha, supported by Tasnia Mahjabin Maliha. Media and coverage was led by Pritom Dutta, while Tazwar Ahmed Seyam focused on system analytics.

Fatima Ashraf is a Campus Ambassador for The Daily Star from Military Institute of Science & Technology (MIST).



BRACU Duburi – Autonomous Underwater Vehicle (AUV)

Role: Team Lead, AI and Machine Vision
Duration: June 2021 – April 2024
Organization: BRAC University Duburi
Project Type: Autonomous Underwater Vehicle (AUV)

Project Overview

BRACU Duburi is a multidisciplinary **Autonomous Underwater Vehicle (AUV)** research team at BRAC University dedicated to advancing underwater robotics through innovation, collaboration, and engineering excellence. **Established in 2017**, the team consists of seven specialized sub-teams covering AI & Autonomous Systems, Sensor & Electronics, Control & Communication, Mechanical Design, 3D Design, Operations, and Outreach.
The team **designs, develops, and deploys autonomous underwater robots** capable of performing complex underwater missions using advanced sensing, navigation, computer vision, and control systems. Through continuous research and international competition participation, BRACU Duburi has established itself as one of **Bangladesh's leading underwater robotics teams** with multiple international recognitions.

Technical Specifications

- **Vehicle Type:** Autonomous Underwater Vehicle (AUV)
- **Main Computer (AI & Vision):** NVIDIA Jetson Orin Nano
- **Vision:** SSD-MobileNet, YOLO, Mask-RCNN, CLAHE, Tracking, TensorRT, Jetson Inference
- **Embedded Controller:** Pixhawk 2.4.8
- **Depth Sensor:** Blue Robotics Bar30 Depth Sensor
- **Inertial Navigation System:** VectorNav VN-200 IMU
- **Velocity Sensor:** Nortek Nucleus 1000 Doppler Velocity Log (DVL)
- **Cameras:** 3 × 1080p Low-Light Cameras
- **Underwater Communication:** Fathom-X Tether Interface with Optical Fiber Networking Unit
- **Propulsion System:** 8 × Blue Robotics T200 Thrusters
- **Power Supply:** 2 × 12 V 10,000 mAh LiPo Batteries
- **Weight:** 27.7 kg (61 lbs)
- **Dimensions:** 21 × 17 × 8 in



Nortek Nucleus 1000



PixHawk 2.4.8



Jetson Orin Nano



Vectornav VN-100

My Responsibilities

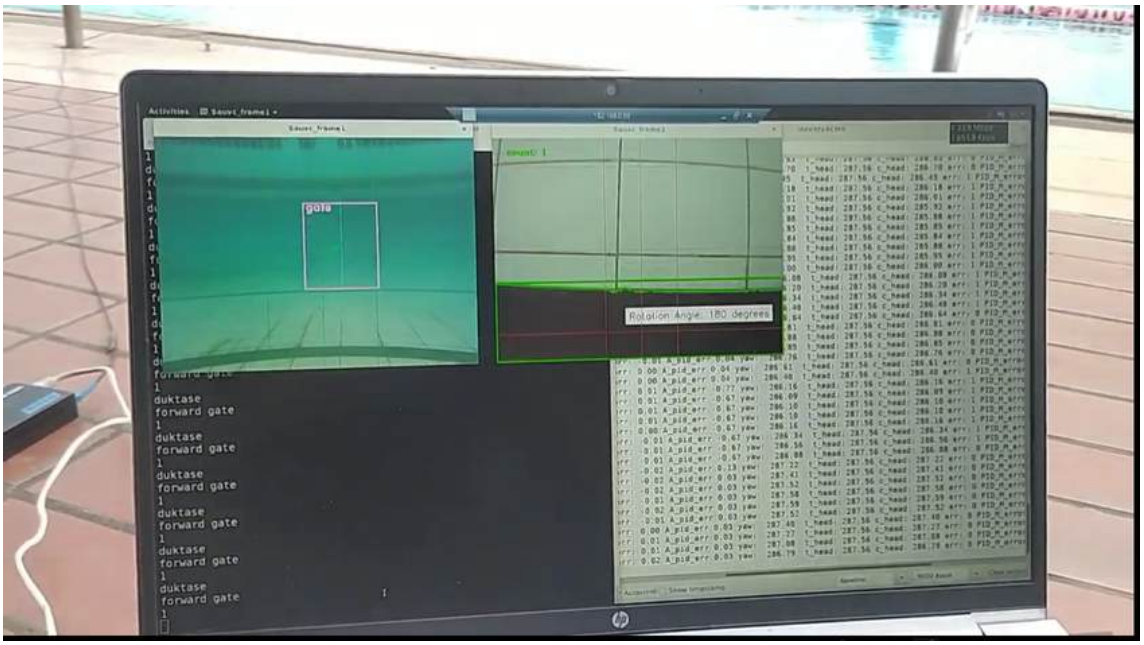
- **Led** the **AI & Machine Vision sub-team** for the development of intelligent perception and autonomous systems.
- **Developed computer vision** algorithms for underwater object detection and visual perception.
- Collected and Preprocessed dataset in **different underwater environments** with different color contrast.
- **Integrated the NVIDIA Jetson Orin Nano** for onboard AI processing and autonomous decision-making.
- **Implemented the Pixhawk-based vehicle control system** and integrated it with the onboard AI computer.
- **Configured and integrated the VectorNav VN-100** Inertial Navigation System (INS) for accurate navigation.
- **Designed and implemented the onboard communication architecture** for seamless subsystem integration.
- Conducted system integration, debugging, sensor calibration, and performance optimization.
- Performed pool testing and field validation to ensure reliable autonomous operation**.**

Project Achievements

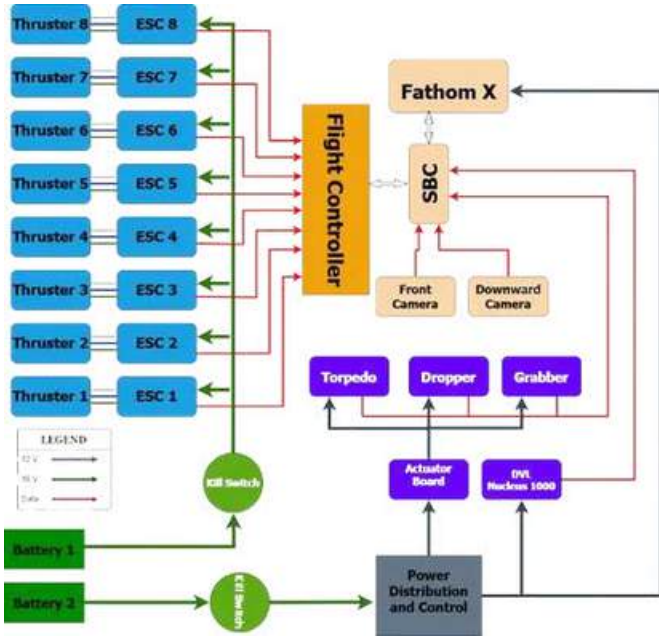
- Achieved 1st Runner-Up at RoboSub 2023 in California, USA.
- Received the Ingenuity Special Award at RoboSub 2023.
- Won 1st Place – Website Design Competition at RoboSub 2024.
- Received the Entrepreneurship Award at RoboSub 2025.
- Recognized as the Best Rookie Team at RoboSub 2022.
- Established BRACU Duburi as one of the leading university underwater robotics teams in Bangladesh through consistent international achievements.

Field Testing & Outreach

- Conducted extensive pool and open-water testing to validate navigation, sensing, propulsion, and control systems.
- Performed iterative hardware and electronics improvements based on real-world testing results.
- Demonstrated the AUV during university exhibitions, technical workshops, and outreach events.
- Promoted underwater robotics research through technical presentations and collaborative engineering activities.



Real-Time Vision System



Circuit Diagram





Duburi v4.0

Photos and Media Coverage



Brac University team ranks second at RoboSub 2023

7 AUGUST 2023, 15:20 PM • UPDATED 7 AUGUST 2023, 22:05 PM • TECH & STARTUP

SHARE

Tech & Startup Desk

During the RoboSub 2023 finals held on August 6 in San Diego, California, US, the Bangladeshi team won the first runners-up position with their autonomous underwater vehicle BRACU Duburi 4.0.

Featured in The Daily Star Newspaper

Click the image to visit the page



তথ্য ও প্রযুক্তি

উদ্ধার করবে ব্র্যাকইউ ডুবুরি







BRAC University celebrates BRACU Duburi's grand success at 'Robosub 2023'



BRAC University organised an event to celebrate BRACU Duburi's grand success at 'Robosub 2023' on 27 August at BRAC University auditorium.

Featured in The Business Standard

Click the image to visit the page



Certificates



Certificate of Participation

Presented to

Shaownak Md. Ibne Shahriar Talukder

Brac University

BRACU Duburi

For successful participation in the

26th Annual International RoboSub Competition

July 31 – August 6, 2023

NIWC Pacific's TRANSDEC – San Diego, California, USA

Lindsey Groark

Vice President of Programs, RoboNation, Inc.

robonation robosub



Certificate of Award

THIS CERTIFICATE ACKNOWLEDGES THAT

BRACU DUBURI

BRACU DUBURI

Brac University

HAS BEEN RECOGNIZED FOR OUTSTANDING ACHIEVEMENT
AT THE 2023 ROBOSUB COMPETITION TO RECEIVE
SECOND PLACE OVERALL - $3,750

robonation Robotics Community August 6, 2023 robosub

BRACU Mongol-Tori – Autonomous Mars Rover

Role: AI & Autonomous Systems Lead
Project Type: Autonomous Mars Rover
Duration: January 2022– August 2023
Organization: BRAC University Mongol-Tori

Project Overview

BRACU Mongol-Tori is BRAC University's multidisciplinary **Mars Rover research** team dedicated to **developing autonomous robotic systems** for **planetary exploration**. The rover was designed to perform autonomous navigation, terrain traversal, object detection, and robotic manipulation while operating in simulated extraterrestrial environments. The project integrates **artificial intelligence, computer vision, robotics, embedded systems, and autonomous navigation** to address challenges encountered in international rover competitions.

Technical Specifications

Platform: Autonomous Mars Rover
Main Computer: NVIDIA Jetson Nano
Embedded Controllers: Raspberry Pi & Arduino
Robotics Framework: Robot Operating System (ROS)
Navigation System: VectorNav Inertial Navigation System (INS)
Perception Sensor: LiDAR
Programming Languages: Python, C++
Computer Vision: Pytorch, TensorFlow, OpenCV, YOLO, SSD, Faster-RCNN, TensorRT
Manipulator: 6-DOF Robotic Arm with Inverse Kynematics
Autonomous Capabilities: Autonomous Navigation, Obstacle Detection, Path Planning & Robotic Manipulation, Object Detection



Arduino Mega



RP LiDAR



Vectornav VN-100



Jetson Orin Nano

My Responsibilities

- **Led** the **AI & Autonomous Systems** sub-team throughout the project.
- Developed **AI and machine vision algorithms** for autonomous perception and components picking.
- Implemented the rover **control system** and **autonomous navigation**.
- Integrated the **VectorNav** INS for localization and navigation.
- Developed the **inverse kinematics of the 6-DOF robotic arm** for autonomous manipulation.
- Integrated ROS, LiDAR, Jetson Nano, Raspberry Pi, and Arduino into a unified robotic system.
- Conducted system integration, testing, debugging, and performance optimization.

Project Achievements

- Successfully represented BRAC University at the **International Rover Challenge** (IRC) in Bengalore, India.
- Participated in the **University Rover Challenge (URC)** in Utah, USA.
- **Champion** in AUST Rover Challenge Bangladesh.

Outreach & Technical Engagement

- Conducted seminars and technical workshops on robotics, autonomous systems, and computer vision.
- Promoted robotics research and innovation through knowledge-sharing sessions and technical demonstrations.
- Mentored junior team members in AI, ROS, autonomous navigation, and robotic system integration.

Photos and Media Coverage













Certificates



Certificate of Participation

Awarded to

Shaownak Md. Ibne Shahriar Talukder

for contribution in

The Mars Society's

2023

University Rover Challenge

As a team member from

BRACU Mongol-Tori

BRAC University

Robert Zubrin, President, The Mars Society

Kevin Sloan, Director, University Rover Challenge

University Rover Challenge (URC)



PRESIDENCY UNIVERSITY

SPACE ROBOTICS SOCIETY

INTERNATIONAL ROVER CHALLENGE 2023

International Rover Challenge

CERTIFICATE OF PARTICIPATION

This certificate is presented to Shaownak Md. Ibne Shahriar of BRAC University, Bangladesh

for participating in the International Rover challenge (IRC),

a space research and innovation competition organized by Space Robotics Society (SPROS)

held at Presidency University, Bengaluru, India from 27th to 31st of January 2023.

DR. D SUBHAKAR
Vice Chancellor,
Presidency University

DR. C.S RAMESH
Dean-Research & Development,
Presidency University

SAGAR DHAKA
Founder, SPROS

JUSTINA PELC
Lead Judge, IRC 2023

Vikasana

Space Robotics Society

International Rover Challenge (IRC)

Shurokkha – Autonomous Rescue Robot

Role: Robotics & AI Engineer
Project Type: Autonomous Search and Rescue Robot

Project Overview

Shurokkha is an autonomous rescue robot developed to assist first responders during natural disasters and hazardous environments. The robot is designed to navigate challenging terrain, detect victims, and deliver essential medical supplies or emergency resources while minimizing risks to rescue personnel. By integrating robotics, artificial intelligence, and computer vision, the platform enhances situational awareness and supports rapid disaster response operations.

Technologies

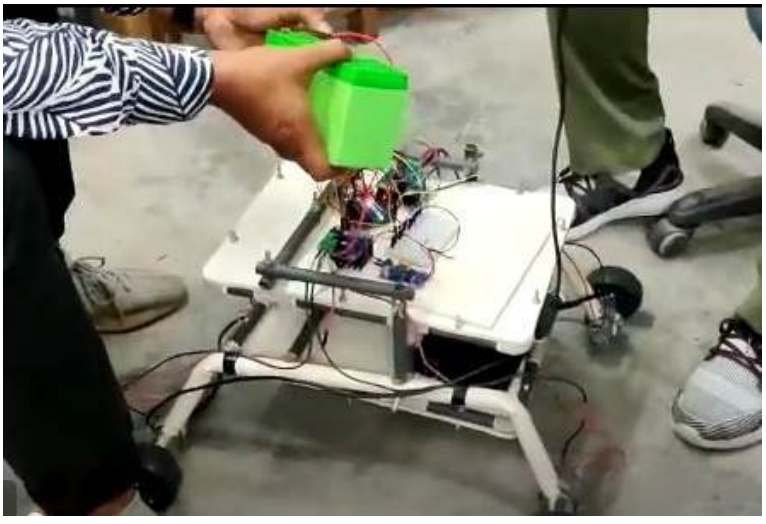
AI & Computer Vision: YOLO, OpenCV, Pytorch, ONNX
Embedded Systems: Raspberry Pi, Arduino
Navigation & Control: Autonomous Navigation, Obstacle Avoidance, Wireless Teleoperation
Sensors: Camera, Ultrasonic Sensors, Environmental Sensors (temperature, humidity)

My Responsibilities

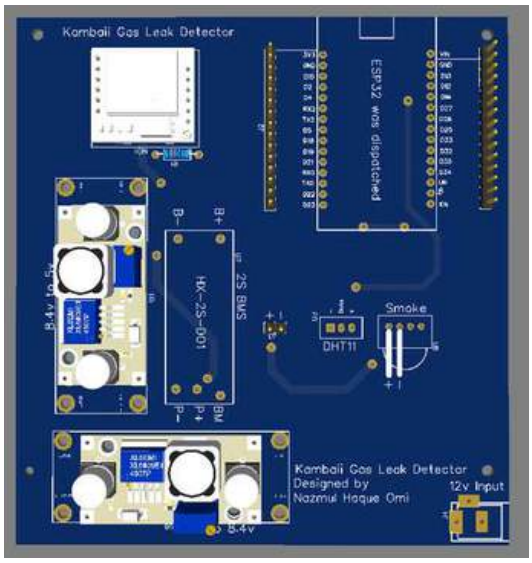
- Designed and developed the AI-based perception system for victim detection.
- Implemented autonomous navigation and obstacle avoidance algorithms.
- Integrated onboard sensors, embedded controllers, and communication modules.
- Developed the robot's software architecture for real-time monitoring and control.
- Conducted system integration, testing, debugging, and field validation.

Project Highlights

- Developed an autonomous robotic platform for disaster response and humanitarian assistance.
- Integrated AI, computer vision, embedded systems, and autonomous navigation into a unified rescue robot.
- Demonstrated autonomous navigation and real-time victim detection in simulated disaster scenarios.
- Designed to support emergency response by improving operational safety and situational awareness.



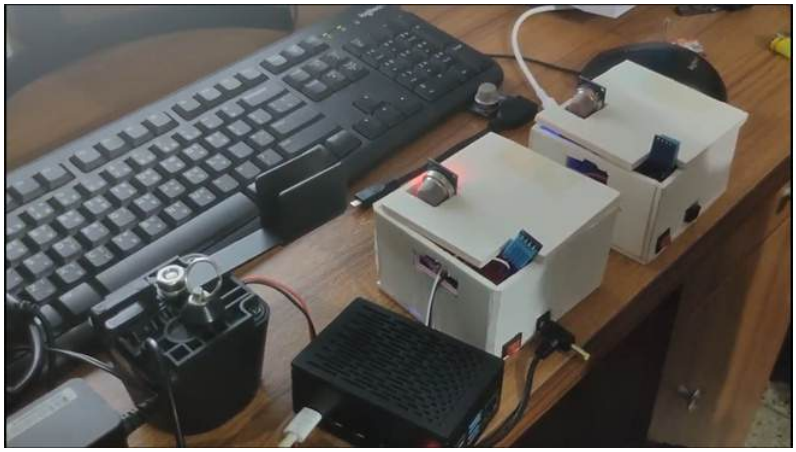
IoT Gas Flow Controller — Distributed Gas Leak Detection System (2023)

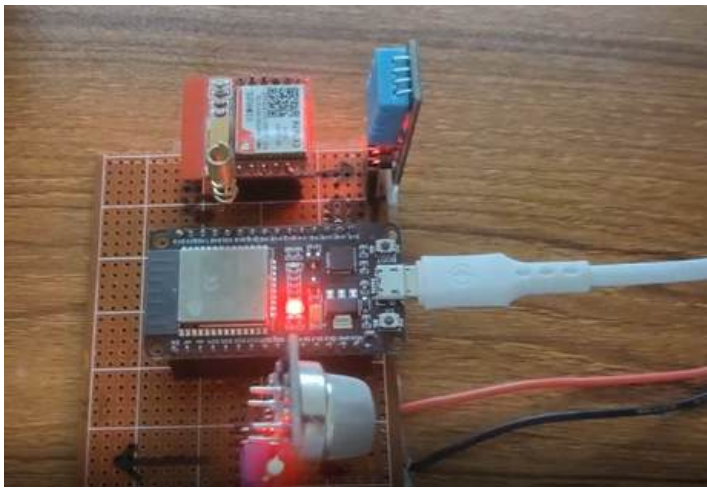


Overview: Developed a distributed gas leak detection and automatic shutoff system using a network of **ESP32** sensor nodes communicating wirelessly via the **MQTT** protocol to a central **Raspberry Pi** controller. Each ESP32 node continuously monitors gas concentration levels and publishes real-time data to the **MQTT broker** running on the Raspberry Pi. Upon detecting a gas leak beyond a safe threshold, the Raspberry Pi instantly triggers an **automated gas flow valve**, preventing potential hazards. The distributed architecture ensures wide area coverage across multiple zones, making the system highly scalable and suitable for both residential and industrial safety applications.

Technology Stack: ESP32, Automated Gas Flow Valve, MQ-2,4,5,6 Gas Sensor, MQTT Broker, GSM Module, Humidity Sensor

Github: https://github.com/Shaownak/Gas-Detection-System







Underwater Image Enhancement using Generative Adversarial Networks (GANs)

Project Type: Underwater Computer Vision
Company: Dubotech Limited
Domain: Deep Learning, Image Processing

Project Overview

Developed an AI-based **underwater image enhancement system** to improve image quality in **low-visibility aquatic environments**. The project evaluated both deep learning and traditional image enhancement techniques to **restore color, improve contrast, and enhance visual clarity** for underwater perception and autonomous robotic applications.

Core Technologies

- **Deep Learning:** FUnIE-GAN
- **Image Processing:** CLAHE, Histogram Equalization, Auto Color Correction
- **Programming:** Python
- **Frameworks:** PyTorch, OpenCV

My Responsibilities

- Implemented and evaluated FUnIE-GAN for underwater image enhancement.
- Developed image enhancement pipelines using CLAHE, Histogram Equalization, and Auto Color Correction.
- Compared enhancement techniques using qualitative and quantitative image analysis.
- Optimized image quality for underwater computer vision applications.

Project Highlights

- AI-powered underwater image enhancement using GANs.
- Improved underwater visibility, color restoration, and image contrast.
- Compared deep learning and conventional enhancement techniques.
- Enhanced visual perception for autonomous underwater robotics and marine inspection.

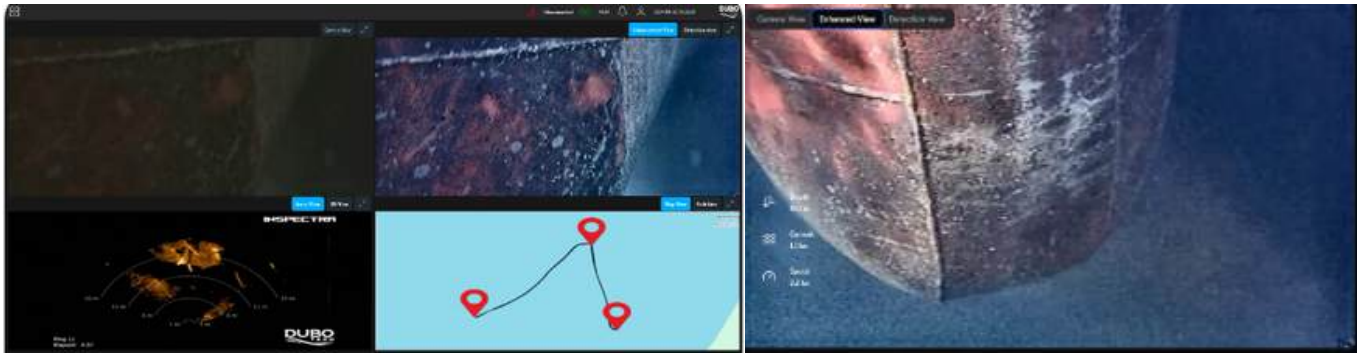


Image Enhancement Dashboard

GLE-SSD-VR: Point Cloud-Based 3D Object Detection for Unmanned Ground Vehicles

Project Type: Undergraduate Thesis
Domain: Robotics, Autonomous Driving, Computer Vision

Project Overview

Designed and developed **GLE-SSD-VR**, a novel **hybrid 3D object detection** framework by integrating the strengths of **GLE-NetVR** and **SE-SSD**. The research aimed to enhance object detection accuracy and robustness from **LiDAR point cloud** data for autonomous systems. The proposed model was trained and evaluated on the **KITTI** Dataset to detect vehicles, pedestrians, and cyclists in complex driving environments, contributing to advanced perception for autonomous robots and intelligent vehicles.

Core Technologies

- **Deep Learning:** PyTorch
- **3D Detection Framework:** GLE-SSD-VR (Proposed Hybrid Model)
- **Base Architectures:** GLE-NetVR, SE-SSD
- **Dataset:** KITTI Dataset
- **Programming Languages:** Python
- **Libraries:** OpenCV, Pytorch
- **Operating Environment:** Windows

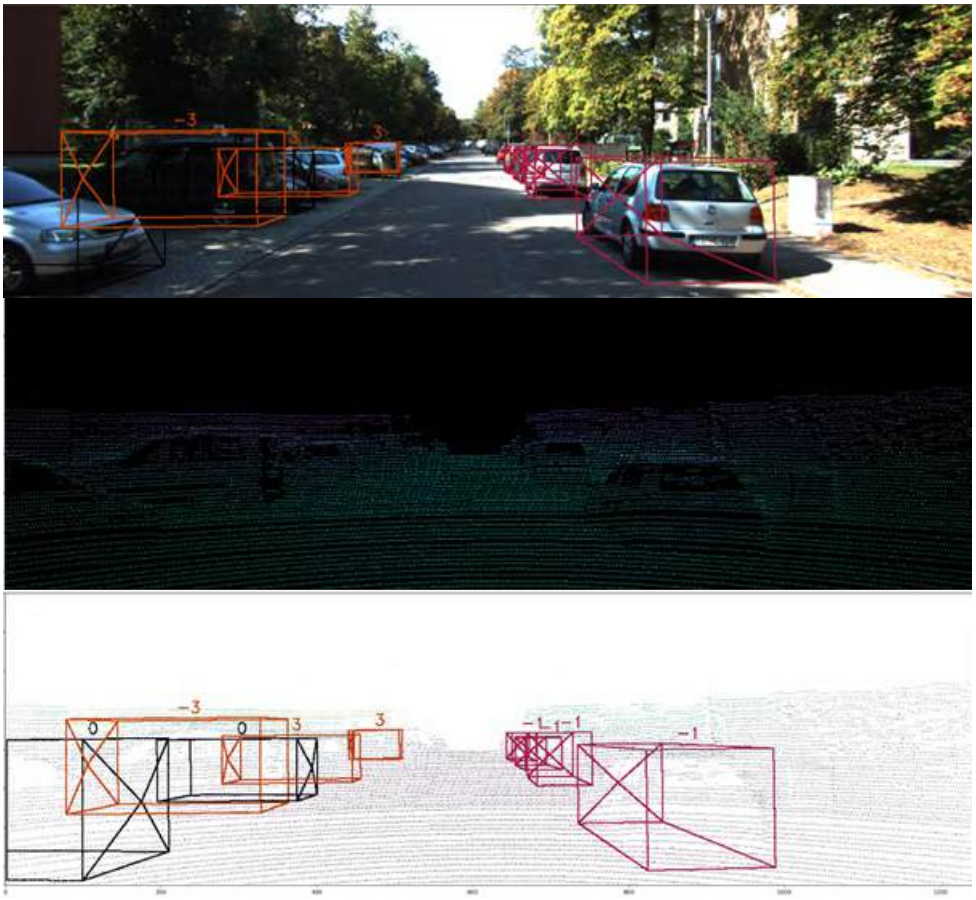
My Responsibilities

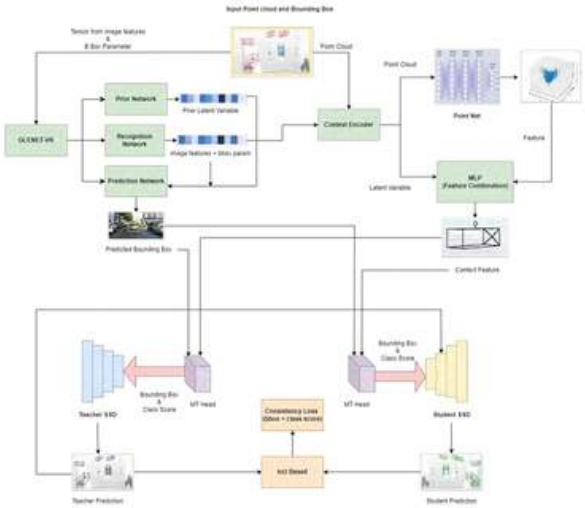
- Designed and implemented the GLE-SSD-VR hybrid architecture by integrating GLE-NetVR and SE-SSD.
- Developed the training and evaluation pipeline for LiDAR-based 3D object detection.
- Processed and analyzed point cloud data for multi-class object detection and localization.
- Trained, optimized, and evaluated the proposed model using the KITTI dataset.
- Conducted performance analysis using standard 3D object detection evaluation metrics and compared the proposed model with baseline architectures.

Paper Link

Research Highlights

- Proposed the **GLE-SSD-VR hybrid model** for LiDAR-based 3D object detection.
- Combined the **feature representation capability of GLE-NetVR with the self-ensembling strategy of SE-SSD**.
- Evaluated the proposed architecture on the KITTI benchmark for detecting vehicles, pedestrians, and cyclists.
- Advanced research in autonomous perception for robotics and intelligent transportation systems.
- Achieved **88% detection accuracy** on the KITTI benchmark dataset.





Architecture Diagram

AI-Based Driver Safety Monitoring System

Project Type: Intelligent Transportation System (ITS)
Domain: Computer Vision, Artificial Intelligence, Embedded Systems

Project Overview

Developed an **AI-powered Driver Safety Monitoring System** designed to enhance road safety through real-time **driver behavior analysis** and **intelligent event** detection. The system leverages **computer vision** and deep learning to continuously monitor the driver's condition, identify **unsafe behaviors**, and generate **instant alerts**, providing an advanced driver assistance solution for accident prevention and intelligent transportation applications.

Core Technologies

- **Computer Vision:** YOLO-Pv2, OpenCV, MediaPipe, MiDAS, Monocular Depth Estimation
- **Programming Languages:** Python
- **Embedded Systems:** Nvidia Jetson Nano, Arduino

My Responsibilities

- Designed and developed the end-to-end AI-based driver monitoring system.
- Implemented **real-time computer vision algorithms** for driver behavior analysis.
- Developed detection modules for driver **drowsiness, distraction, and unsafe driving** activities.
- Integrated **embedded hardware, onboard cameras, and alert mechanisms** into a unified safety platform.
- Optimized the system for reliable real-time performance and low-latency inference.

Key Features

- Real-time **driver monitoring** and behavior analysis
- **Driver drowsiness** and fatigue detection
- **Eye closure and yawning** detection
- Driver **distraction** monitoring
- **Mobile phone usage** detection
- **Car Lane and Distance** detection
- Real-time visual and audible safety alerts
- Embedded AI deployment for continuous monitoring

Project Highlights

- Developed an intelligent AI-driven Driver Monitoring System for real-time road safety applications.
- Successfully integrated computer vision, embedded systems, and deep learning into a unified ADAS prototype.
- Demonstrated reliable real-time detection of multiple driver safety events for proactive accident prevention.
- Showcased the practical application of AI and embedded vision technologies in intelligent transportation systems.

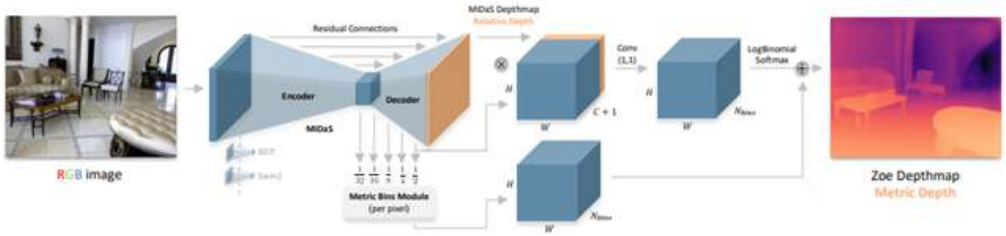
Github: https://github.com/Shaownak/Driver-Safety-System

Featured Photos

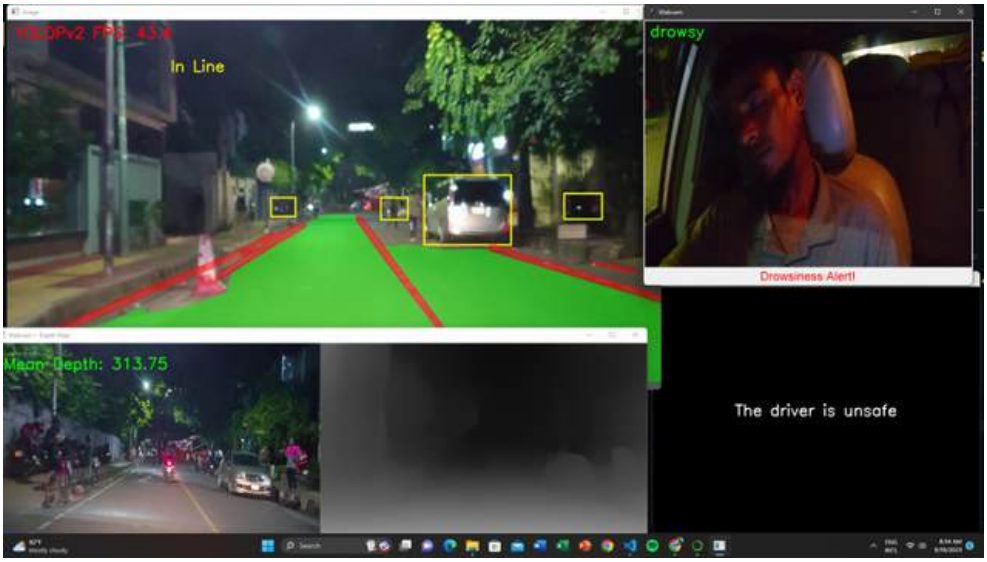




Measuring car distance with depth estimation



Architecture Diagram



Whole System in Real Time

Geospatial Traffic Optimization System

Project Type: Geospatial AI & Intelligent Transportation System
Domain: GIS, Machine Learning, Traffic Analytics

Project Overview

Developed an **AI-powered traffic prediction** and r**oute optimization** system for **Dhaka's road network** using spatial-temporal machine learning. The system analyzes traffic patterns to optimize routing efficiency and support **intelligent transportation** and **supply chain logistics.**

Core Technologies

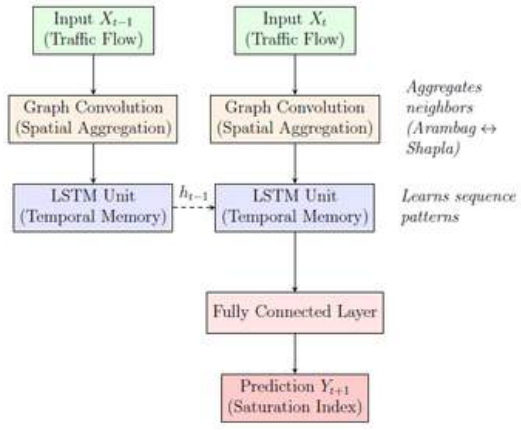
AI/ML: Graph Neural Networks (GNNs), LSTM, XGBoost
GIS: OSMnx, Folium
Data Analysis: Pandas, NumPy

My Responsibilities

- Developed **spatial-temporal traffic prediction** models using **GNNs** and **LSTM.**
- Implemented **XGBoost-based** ensemble learning for improved prediction accuracy.
- Integrated **GIS road network** data for intelligent route optimization.
- Built interactive **traffic visualization and analytics dashboards**.

Project Highlights

- AI-driven traffic prediction and route optimization.
- Integrated GNNs, LSTM, and XGBoost into a unified prediction framework.
- Applied Geospatial AI to improve urban mobility and logistics planning.



Architecture Diagram



Real Time Traffic Congestion Detection

AI-Based Drone Detection & Tracking System

Project Type: Intelligent Surveillance System
Domain: Computer Vision, Deep Learning

Project Overview

Developed an AI-powered drone **detection** and **tracking** system for real-time aerial **surveillance**. The system utilizes deep learning and computer vision to detect, identify, and continuously track UAVs from another UAV in real time.

Core Technologies

- **Computer Vision:** Fine Tuned YOLOv11, DeepSORT, ONNX, OpenCV, Pytorch
- **Hardware:** Raspberry Pi 4B

My Responsibilities

- Developed the real-time drone detection and tracking pipeline.
- Integrated **YOLOv8** with **DeepSORT** for continuous object tracking.
- Optimized inference for real-time video processing.

Project Highlights

- Real-time drone detection and tracking
- AI-powered aerial surveillance
- Low-latency video analytics
- Security and defense applications

Github: https://github.com/Shaownak/Drone-Detection-by-fine-tuning-Yolo-v11

MPU6050 Sensor Fusion using Kalman Filter

Project Type: Embedded Robotics & Sensor Fusion
Domain: Robotics, State Estimation, Embedded Systems

Project Overview

Developed a real-time **sensor fusion** system using the **MPU6050 IMU** and a **Kalman Filter** to accurately estimate **device orientation**. The project combines **accelerometer and gyroscope** data to minimize sensor noise and drift, providing stable roll and pitch estimation for robotics and autonomous navigation applications.

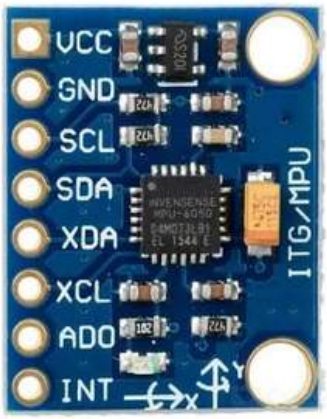
Core Technologies

- **Algorithms:** Kalman Filter
- **Sensors:** MPU6050 (Accelerometer & Gyroscope)
- **Programming:** C++, Arduino
- **Platform:** Arduino

My Responsibilities

- Implemented the Kalman Filter for real-time sensor fusion.
- Integrated accelerometer and gyroscope data for accurate orientation estimation.
- Optimized filtering performance to reduce noise and sensor drift.
- Project Highlights
- Real-time roll and pitch estimation
- Robust sensor fusion using Kalman Filter
- Improved orientation accuracy and stability
- Applicable to UAVs, mobile robots, and autonomous systems

Github: https://github.com/Shaownak/MPU6050-Sensor-Fusion-using-Kalman-Filter



MPU6050 3 axis IMU

Vision-Based Object Detection & Autonomous Navigation

Project Type: Computer Vision & Mobile Robotics
Domain: Autonomous Navigation, Embedded AI

Project Overview

Developed a **vision-based autonomous navigation** system that detects objects in real time and enables a mobile robot to **navigate** accordingly. The system integrates **computer vision** with **Arduino-based motion control** for intelligent obstacle awareness and autonomous movement.

Core Technologies

- **AI & Vision:** OpenCV, Pytorch, Yolo-v3
- **Programming:** Python, C++
- **Embedded Platform:** Arduino
- **Communication:** Serial UART

My Responsibilities

- Developed the real-time object detection pipeline using OpenCV.
- Integrated computer vision with Arduino for autonomous robot control.
- Implemented serial communication between the vision system and embedded controller.
- Tested and optimized navigation performance in real-world scenarios.

Project Highlights

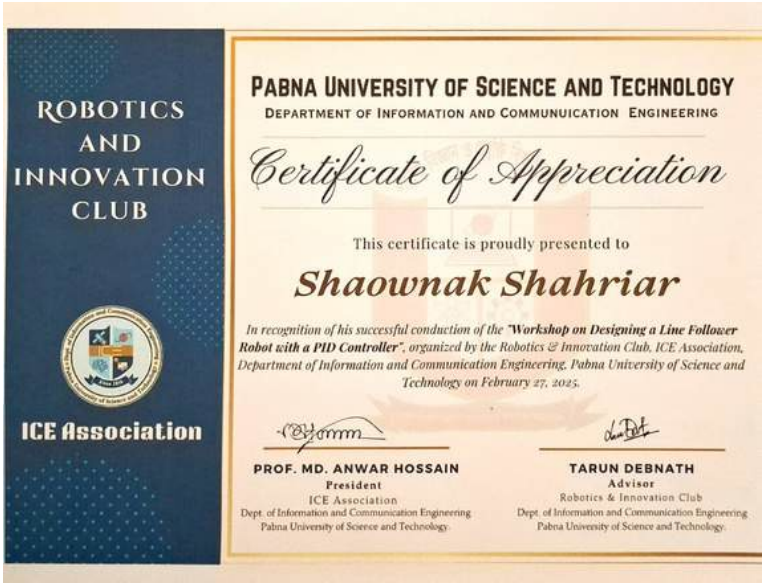
- Real-time object detection and autonomous navigation
- Vision-guided robot control
- Embedded AI and hardware integration
- Intelligent obstacle-aware navigation

Github: https://github.com/Shaownak/Object-Detection-and-Navigation-with-Computer-Vision-and-Arduino

Other Certifiicates



Kibo Robot Programming Competition



Guest Robotics Instructor (Pabna University of Science and Technology)



Robotics Instructor (Brac University)



Best Thesis Award

Notable Online Certifications



Deep Learning Specialization by deeplearning.ai



HackerRank Certified Software Engineer



HackerRank SQL Advanced Certification

Conclusion

Throughout my academic and professional journey, I have been driven by a passion for developing intelligent robotic systems that combine artificial intelligence, computer vision, autonomous navigation, and embedded technologies to solve real-world challenges. From autonomous aerial, underwater, and ground robots to enterprise AI and advanced perception systems, each project has strengthened my expertise in designing, integrating, and deploying intelligent autonomous solutions.
I look forward to contributing my skills, creativity, and engineering mindset to developing next-generation robotic systems that create meaningful impact in industry and society.

Contact

Shaownak Shahriar
Robotics Engineer | Computer Vision Engineer | AI Researcher

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💻 **GitHub:** github.com/Shaownak
🌐 **Portfolio:** shaownak.github.io